PCAP File Analysis Scenario – By Lior Tzlil Friedman:

Part A: Investigation of Suspected Activity:

1. Identify recurring patterns in TCP traffic that may indicate an attempted attack.

We can identify an attempted attack because of we reserve a lot raptly of TCP requests from the same IP address – 10.55.61.205 , on a **close port 80**

31164 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443094994	…5615
31165 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443095280	…5615
31168 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443095345	…5615
31170 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443095392	…5615
31172 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443095464	…5615
31173 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443095522	…5615
31174 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443095591	…5615
31175 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.443095659	…5615

31156 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441943863	…5614
31157 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441944076	…5614
31185 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441944144	…5614
31158 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441944216	…5614
31160 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441944289	…5614
31161 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441944352	…5614
31162 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441944414	…5614
31163 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused] 60	TCP	10.55.61.158	10.55.61.205	407.441944478	…5614

2. How many distinct IP addresses are involved in initiating the traffic?

We have only 2 distinct IP addresses , Destination - 10.55.61.158 ,
Source - 10.55.61.205

ip.src

Info	Length	Protocol	Destination	Source	Time	No
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43827 → 80	58	TCP	10.55.61.205	10.55.61.158	422.101764926	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43828 → 80	58	TCP	10.55.61.205	10.55.61.158	422.101866358	…8150
43829 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101965726	…8150
43830 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101966304	…8150
43831 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101966373	…8150
43832 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101966419	…8150
43833 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101966477	…8150
43834 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101966547	…8150
43835 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101966596	…8150
43836 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	422.101966662	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43829 → 80	58	TCP	10.55.61.205	10.55.61.158	422.102015803	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43830 → 80	58	TCP	10.55.61.205	10.55.61.158	422.102120047	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43831 → 80	58	TCP	10.55.61.205	10.55.61.158	422.102227514	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43832 → 80	58	TCP	10.55.61.205	10.55.61.158	422.102306499	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43833 → 80	58	TCP	10.55.61.205	10.55.61.158	422.102382560	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43834 → 80	58	TCP	10.55.61.205	10.55.61.158	422.102459098	…8150
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 43835 → 80	58	TCP	10.55.61.205	10.55.61.158	422.102536854	…8150

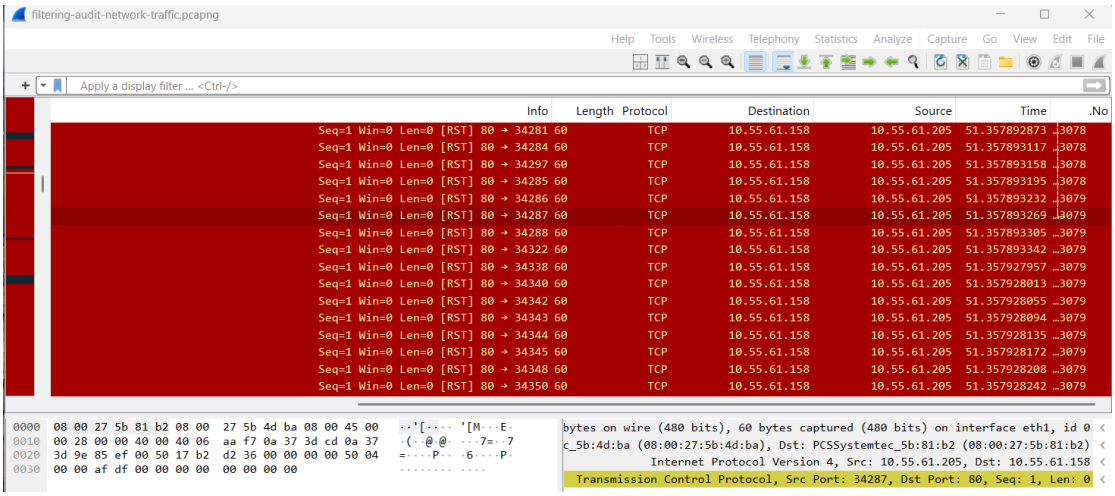
3. Does the target server respond to the incoming traffic, or are the requests ignored?

The target server 10.55.61.158 is ignore to the many requests of TCP SYN , And not receives back a SYN-ACK response

tcp.flags.syn == 1 and tcp.flags.ack == 0

Info	Length	Protocol	Destination	Source	Time	
31177 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438934810	…5614
31128 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438935309	…5614
31130 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438935363	…5614
31131 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438935417	…5614
31134 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438935498	…5614
31136 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438935580	…5614
31137 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438935645	…5614
31138 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.438935721	…5614
31139 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439918835	…5614
31140 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439919109	…5614
31141 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439919163	…5614
31142 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439919207	…5614
31143 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439919257	…5614
31144 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439919314	…5614
31145 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439919361	…5614
31146 → 80 [SYN] Seq=0 Win=512 Len=0 [TCP Port numbers reused]	60	TCP	10.55.61.158	10.55.61.205	407.439919411	…5614

4. What are the specific destination addresses and ports involved ?
The s specific destination addresses and ports on the attack in only
The **IP address 10.55.61.158 and port 80**.



5. Based on the timing and volume of traffic, how would you characterize the intent of this activity?

The Traffic shows us that there is a lot of raptly TCP SYN packets request to **only one** IP address 10.55.61.158 (from IP 10.55.61.205) and only one port 80 , And there is not SYN-ACR or ACK response in a short time between packets.
So we can easily tell we have attack to overwhelm the server with many requests to crash our server.

Part B - Investigation of Brute Force Attempts

1. Locate and isolate traffic related to login attempts.

SSH Attempts

I used the filter: tcp.flags.syn == 1 and tcp.port == 22 , Show only TCP SYN packets related to SSH

tcp.flags.syn == 1 - checks if the **SYN flag** in the TCP packet is set to **1** (true) , So this part filters for connection attempts.

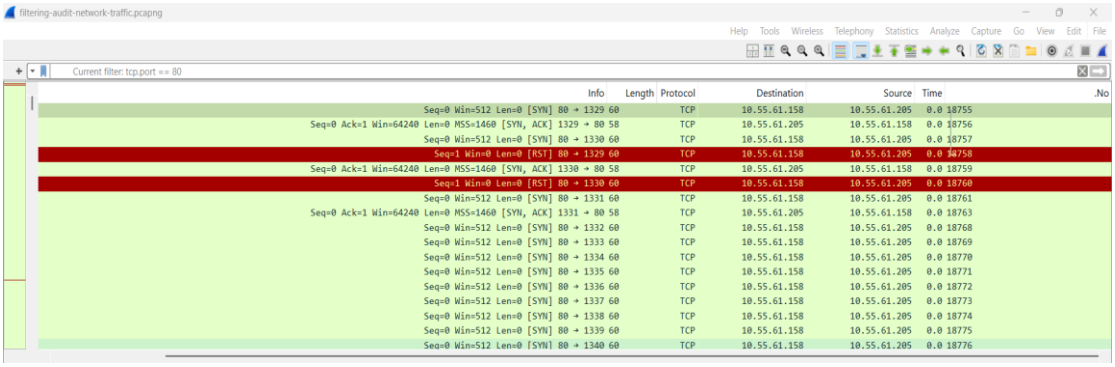
tcp.port == 22 – Port 22 is a default port for SSH connection.

tcp.flags.syn == 1 and tcp.port == 22

Info	Length	Protocol	Destination	Source	Time	No
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570756281 TSecr=0 WS=128 [SYN] 22 → 54536	74	TCP	10.55.61.158	10.55.61.205	0.0	1164239
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570878733 TSecr=0 WS=128 [SYN] 22 → 40294	74	TCP	10.55.61.158	10.55.61.205	0.0	1165495
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 22 → 80	58	TCP	10.55.61.205	10.55.61.158	0.0	1095675
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570878725 TSecr=0 WS=128 [SYN] 22 → 40188	74	TCP	10.55.61.158	10.55.61.205	0.0	1165447
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570672053 TSecr=0 WS=128 [SYN] 22 → 35054	74	TCP	10.55.61.158	10.55.61.205	0.0	312312
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570756280 TSecr=0 WS=128 [SYN] 22 → 54522	74	TCP	10.55.61.158	10.55.61.205	0.0	1164229
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570756282 TSecr=0 WS=128 [SYN] 22 → 54552	74	TCP	10.55.61.158	10.55.61.205	0.0	1164245
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 22 → 80	58	TCP	10.55.61.205	10.55.61.158	0.0	739775
Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 [SYN, ACK] 22 → 80	58	TCP	10.55.61.205	10.55.61.158	0.0	104075
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570756283 TSecr=0 WS=128 [SYN] 22 → 54560	74	TCP	10.55.61.158	10.55.61.205	0.0	1164252
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570878732 TSecr=0 WS=128 [SYN] 22 → 40276	74	TCP	10.55.61.158	10.55.61.205	0.0	1165488
[SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570878734 TSecr=0 WS=128 [TCP Port numbers reused]	74	TCP	10.55.61.158	10.55.61.205	0.0	1165502
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570878727 TSecr=0 WS=128 [SYN] 22 → 40234	74	TCP	10.55.61.158	10.55.61.205	0.0	1165470
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570756278 TSecr=0 WS=128 [SYN] 22 → 54462	74	TCP	10.55.61.158	10.55.61.205	0.0	1164211
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570756286 TSecr=0 WS=128 [SYN] 22 → 54600	74	TCP	10.55.61.158	10.55.61.205	0.0	1164283
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570878731 TSecr=0 WS=128 [SYN] 22 → 40262	74	TCP	10.55.61.158	10.55.61.205	0.0	1165483
Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3570790388 TSecr=0 WS=128 [SYN] 22 → 40322	74	TCP	10.55.61.158	10.55.61.205	0.0	1165267

We have a lot of SYN (it's a flag in TCP protocol , a gives us a Indocin of starting a connection between two end devices) attempts connection , but we don't see a full connection (SYN → SYN-ACK → ACK) , but this indict us on a atomic connoting attempts.

HTTP & FTP Login Attempts

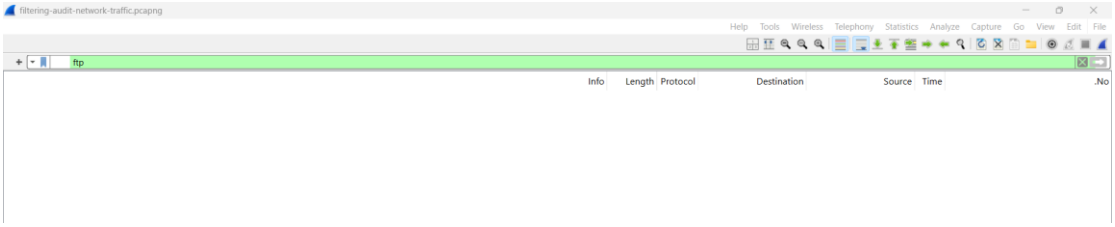


As we can see it does show that we have a HTTP traffic but it not necessarily mean that we have a login attempts, so we need more filtering

Filter - http.request.method == "POST"
Post – a request of user and password

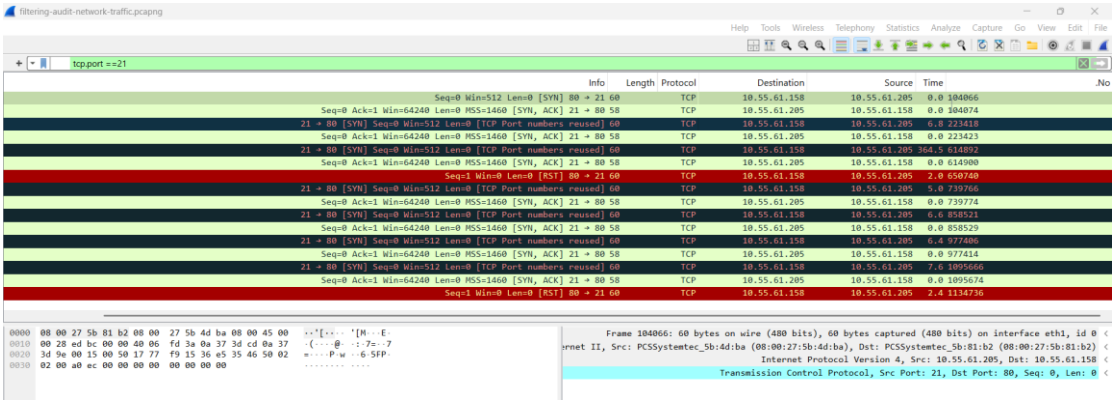
So we can see that we don't have a HTTP login attempts

Filter - ftp



So we can see that we don't have a FTP protocol at all.

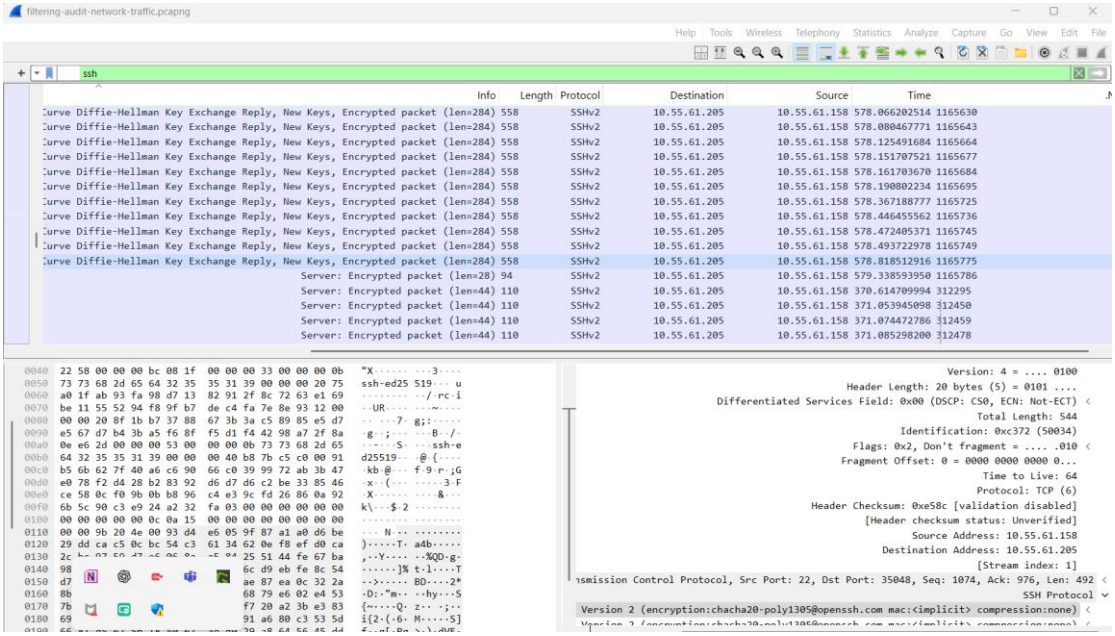
tcp.port ==21



What we see in here is that we do have a few successful communication (SYN-ACK) and we also can see that we mostly getting a rejected or failed do to the RST and also they mark in color red , but we don't see any pacts with USER and PASS.
So we don't see a clear login attempt.

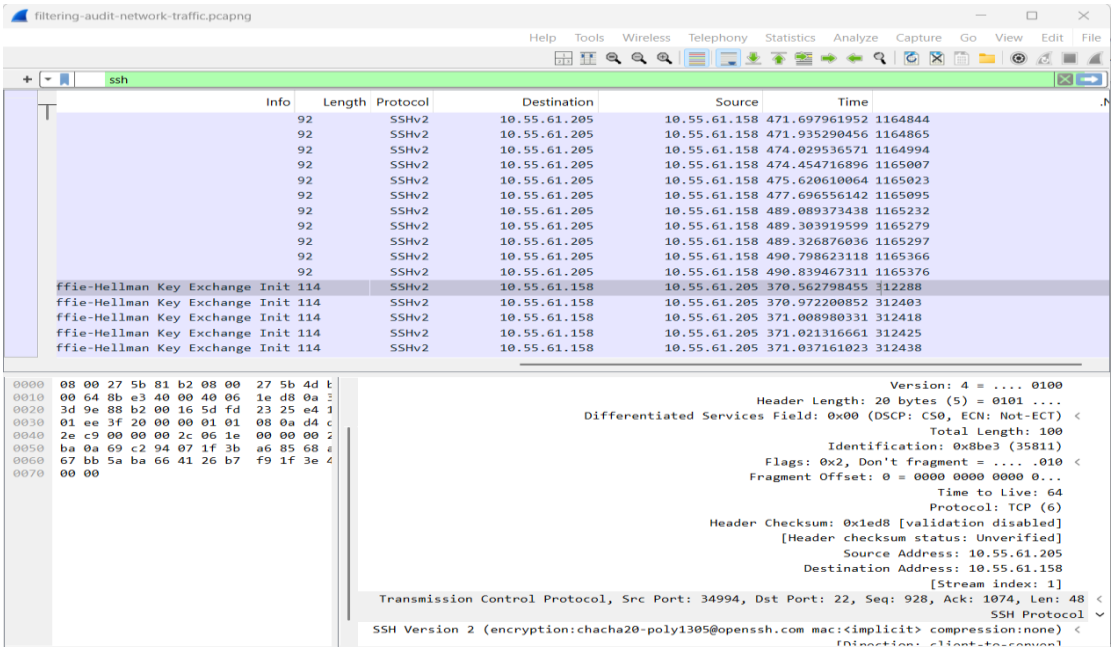
2. Do the attempts originate from a single IP address or multiple sources?

The attempts originate form single IP address **10.55.61.205**



3. Estimate the number of login attempts observed within the capture

I Can't give a exact number or estimate do to the many SSH attempts



4. Was any login attempt seemingly successful? Justify your answer using evidence from the packet data.

There are not any login attempt successfully , I Give all the evidence in question **B.1**

5. Identify any repeating patterns in usernames, request timing, or connection behavior.

As we saw in all of the sections of the question **B** , We got lot of repeating patterns in SSH traffic all the connection attempts was from the same IP address **10.55.61.205** and the packets were sent in milliseconds (which indicates the use of an automated brute force script) and each attempt reused and also no username attempt detected in other words no login attempts as found.

Part C: Comparative Analysis

1. Summarize the key differences between the DDoS and Brute Force attacks observed in the capture.

Feature	Brute Force Attack	DDos Attack
Objective	Gain unauthorized access by guessing passwords	Overload the server and disrupt service
Traffic Behavior	Repeated connection attempts with login intent	Many SYN packets without completing handshake
Primary Port	22	80
Protocol	SSH	HTTP (SYN Flood)
Timing Pattern	Fast and consistent timing, but less obvious	High-volume spikes
Detectability	Harder to detect (looks like normal logins)	Easier to spot (no server response, high traffic)
Detection Indicators	Repeating patterns, fixed IP, short time gaps	SYN without ACK, large volume of traffic
Encrypted Protocols	Yes (SSH hides login info)	No

2. Which indicators helped you identify each type of attack?

The indicator is Traffic Behavior in B.F Attack is the many connection attempts , and in DDos Attack is Many SYN packets without completing handshake.

3. Which of the two attacks was more subtle or difficult to detect? Explain your reasoning.

The Attack that is more difficult to detect is the Brute Force Attack do to SSH encrypted Protocol and the encrypted login attempts so we cannot tell immediately if the login attempted is from inside the organization or unauthorized /outsiders attack.